The *Solution* of a problem is the method of construction which accomplishes the required end.

The *Demonstration* is the proof, in the case of a theorem, that the conclusion follows from the hypothesis; and in the case of a problem, that the construction accomplishes the object proposed.

The *Enunciation* of a problem consists of two parts, namely, the *data*, or things supposed to be given, and the *quaesita*, or things required to be done.

Postulates are the elements of geometrical construction, and occupy the same relation with respect to problems as axioms do to theorems.

A *Corollary* is an inference or deduction from a proposition.

A *Lemma* is an auxiliary proposition required in the demonstration of a principal proposition.

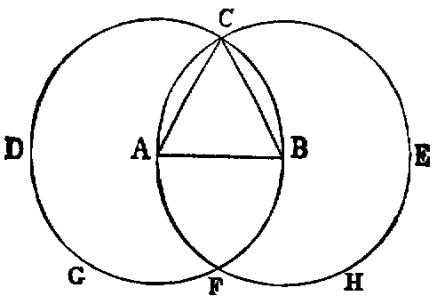
A *Secant* or *Transversal* is a line which cuts a system of lines, a circle, or any other geometrical figure.

Congruent figures are those that can be made to coincide by superposition. They agree in shape and size, but differ in position. Hence it follows, by Axiom VIII., that corresponding parts or portions of congruent figures are congruent, and that congruent figures are equal in every respect.

Rule of Identity.—Under this name the following principle will be sometimes referred to:—"If there is but one X and one Y, then, from the fact that X is Y, it necessarily follows that Y is X."—SYLLABUS.

PROP. I.—PROBLEM.

On a given finite right line (AB) *to construct an equilateral triangle.*



Sol.—With A as centre, and AB as radius, describe the circle BCD (Post. III.). With B as centre, and BA as radius, describe the circle ACE, cutting the former circle in C. Join CA, CB (Post. I.). Then *ABC is the equilateral triangle required.*

Dem.—Because A is the centre of the circle BCD, AC is equal to AB (Def. XXXII.). Again, because B is the centre of the circle ACE, BC is equal to BA. Hence we have proved.

$$AC = AB,$$

and

$$BC = AB.$$

But things which are equal to the same are equal to one another (Axiom I.); therefore AC is equal to BC; therefore the three lines AB, BC, CA are equal to one another. Hence the triangle ABC is equilateral (Def. XXI.); and it is described on the given line AB, *which was required to be done.*